

PIYUSH KUMAR

Software & AI Engineer | Full-Stack Development • RAG Systems • Agentic AI

New Town, Kolkata, India

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Education

B.P. Mandal College of Engineering – Madhepura, Bihar

Aug 2019 – Aug 2023

B.Tech in Electrical and Electronics Engineering

CGPA: 8.04 / 10

Experience

IB Arts Pvt Ltd

Jan 2024 – Present

Software Engineer

Kolkata, West Bengal

- Architected and deployed 6+ production-grade systems, including **multi-tenant SaaS platforms**, **cross-platform interactive apps**, and **automated coding evaluation engines** for Data Science & AI workflows; maintained a 100% on-time delivery rate using **TypeScript, Node.js, and Python**.
- Optimized full-stack performance by refining **Express.js** middleware and database queries, resulting in a **25% reduction in API latency** and a 15% increase in user retention.
- Honors:** Received the **"Creative Master"** award for driving innovative design-thinking solutions and bridging the gap between complex technical requirements and user-centric UI/UX.

Projects

Pensieri - AI Content Ecosystem | *Python, RAG, LangChain, Next.js, Spring Boot, Docker*

Dec 2025 – Present

- Architected a scalable, AI-enhanced blogging platform built with Next.js, Node.js, Spring Boot, and MongoDB, actively supporting over 1,000+ monthly readers and 10+ published writers.
- Engineered an enterprise **RAG (Retrieval-Augmented Generation)** pipeline using vector embeddings and semantic search, enabling context-aware document synthesis and Q&A over knowledge bases.
- Implemented an **Agentic AI** workflow utilizing autonomous tool selection, dynamic function calling, and a **Multi-LLM Orchestration** engine with failover routing across Groq/Llama-3, OpenAI, and Anthropic.
- Engineered Write Pad, a block-based editor featuring a custom-built **Tiptap/ProseMirror** core with integrated AI copilot assistants, drag-and-drop image optimization, and real-time contextual suggestions.
- Engineered a high-performance data pipeline featuring **multi-threaded Java background workers** and a dedicated **Node.js microservice** for SMTP operations and Redis caching.

r-datetime - Headless UI Library (NPM package) | *React, TypeScript, NPM, Rollup.js*

Dec 2025 – Jan 2026

- Published a **zero-dependency**, high-performance React utility for complex date-time selection; optimized the build to a highly-minimized **3KB gzipped** footprint using Rollup.js.
- Engineered an enterprise-ready feature set including **IANA Timezone support**, internationalization (Intl API), and a range-selection engine with customizable constraints (min/max/disabled dates).
- Achieved a peak of **662 weekly downloads** on NPM; implemented strict **TypeScript Generics** and Tailwind-based dynamic theming to provide a flexible, type-safe Developer Experience (DX).

Storezee - On-Demand Logistics App | *React Native, Google Maps SDK, Razorpay, Zustand*

Aug 2025 – Oct 2025

- Built a multi-sided marketplace for luggage storage; integrated **Google Maps SDK** for real-time proximity searching for 50+ storage hubs.
- Developed an event-driven status update system for real-time tracking of luggage transit, synchronizing live states between users and store representatives to ensure end-to-end transparency.

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), Java, C/C++, HTML5, CSS3, SQL

AI & ML Engine: RAG Pipelines, Agentic AI, LangChain, LangGraph, MongoDB Atlas Vector Search, Embeddings, Function Calling, Prompt Engineering, Fine-tuning (LoRA), OpenAI / Anthropic / Gemini APIs, Groq, Hugging Face

Backend & DB: Node.js, Express.js, Spring Boot, MongoDB, Redis, Firebase, Supabase, REST APIs, Microservices

Frontend: React.js, Next.js, React Native (Expo/CLI), Redux, Zustand, Tailwind CSS, Shadcn UI, Jest

DevOps & Tooling: Docker, AWS, MLOps (LangSmith), Git, GitHub Actions, Postman, Rollup.js, Figma